



Modelling Financial Markets

Chapter 11: Continuous-Time Models



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Acronyms used in this material (1/2)

- ▣ ACF: AutoCorrelation Function
- ▣ AI: Artificial Intelligence
- ▣ AR: AutoRegressive (model); in event studies: Abnormal Return
- ▣ ARFIMA: AutoRegressive Fractionally Integrated Moving Average (model)
- ▣ ARMA: AutoRegressive Moving Average (model)
- ▣ BET: Bucharest Exchange Trading (index)
- ▣ BSS: Brownian SemiStationary (process)
- ▣ BVB: Bursa de Valori București (the Bucharest Stock Exchange)
- ▣ CBOE: Chicago Board Options Exchange
- ▣ CI: Confidence Interval
- ▣ CIR: Cox–Ingersoll–Ross (model)
- ▣ CKLS: Chan–Karolyi–Longstaff–Sanders (model)
- ▣ CLT: Central Limit Theorem
- ▣ DGS10: 10-Year Treasury Constant Maturity Rate (FRED series code)
- ▣ DTB3: 3-Month Treasury Bill: secondary market rate (FRED series code)
- ▣ EWMA: Exponentially Weighted Moving Average
- ▣ FRED: Federal Reserve Economic Data
- ▣ GARCH: Generalised AutoRegressive Conditional Heteroskedasticity
- ▣ GBM: Geometric Brownian Motion
- ▣ HAC: Heteroskedasticity and Autocorrelation Consistent (standard errors)
- ▣ JB: Jarque–Bera (test)
- ▣ LLM: Large Language Model
- ▣ LR: Likelihood Ratio (test)
- ▣ MCS: Model Confidence Set
- ▣ MLE: Maximum Likelihood Estimation
- ▣ MSE: Mean Squared Error

Acronyms used in this material (2/2)

- ▣ **NLLS: Non-Linear Least Squares**
- ▣ **OLS: Ordinary Least Squares**
- ▣ **OU: Ornstein–Uhlenbeck (process)**
- ▣ **PDE: Partial Differential Equation**
- ▣ **QLIKE: Quasi-LIKelihood (loss function)**
- ▣ **QQ: Quantile–Quantile (plot)**
- ▣ **RFSV: Rough Fractional Stochastic Volatility (model)**
- ▣ **SDE: Stochastic Differential Equation**
- ▣ **SVJ: Stochastic Volatility with Jumps (model)**
- ▣ **US: United States**
- ▣ **VIX: CBOE Volatility Index**
- ▣ **XFG: Quantlet prefix of Applied Quantitative Finance (Härdle, Chen & Overbeck)**

Chapter Route and Outcomes

The question and the route

- ▣ Prices are recorded daily or every few milliseconds: can one model describe them at **every** time scale at once?
- ▣ Part I: Brownian motion, Itô calculus, simulation, geometric Brownian motion, mean reversion
- ▣ Part II: estimating diffusions, change of measure, jumps, stochastic volatility, models against data, a case study, AI-assisted discovery
- ▣ History of the field and technical detail: [▶ Appendix](#), before the references

After this chapter you can

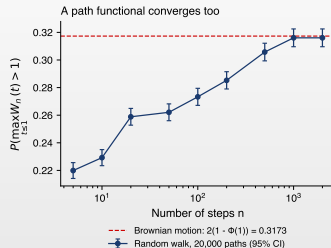
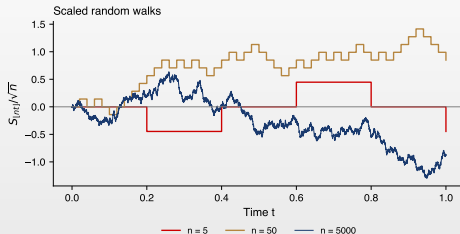
- ▣ apply Itô's lemma and simulate a stochastic differential equation with a known error order
- ▣ estimate GBM, Ornstein–Uhlenbeck, Merton and Heston on data, with standard errors that respect what the sample identifies
- ▣ say which feature of the data each model reproduces, and which feature none of them does
- ▣ Option pricing ([Chapter 12](#)), interest-rate models and risk management are written in this language

Data Used in This Chapter

Series	Type	Period	Days
S&P 500	US equity index	1990–2026	9 245
BET	BVB equity index	2000–2026	6 670
Bitcoin	crypto-asset, 7 days a week	2014–2026	4 384
VIX	CBOE implied volatility index of the S&P 500	1990–2026	daily
DTB3	3-month US Treasury bill rate (FRED)	1954–2026	daily

- Daily closing values up to 18 September 2026; daily log returns $r_t = \ln P_t - \ln P_{t-1}$, each series on its own calendar (cleaning rules: [Appendix](#))
- Time in years, daily step $\Delta t = 1/252$ (Bitcoin: $1/365$); drifts and volatilities annualised
- W_t : standard Brownian motion; S_t : price; v_t : instantaneous variance; $\mathcal{F}_t$: the information (filtration) up to t

From Random Walk to Brownian Motion: Donsker's Theorem



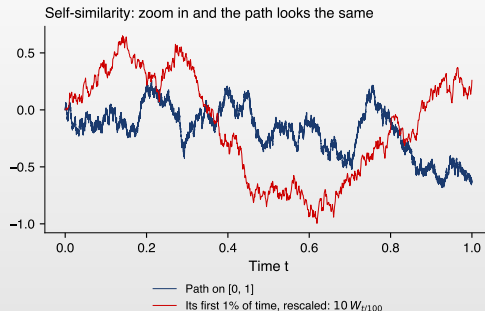
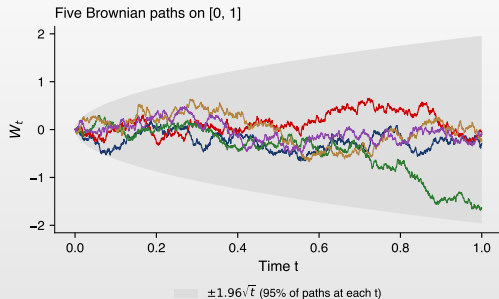
- Scaled random walk $W_n(t) = S_{[nt]}/\sqrt{n}$, $S_k = \xi_1 + \dots + \xi_k$, steps i.i.d. with mean 0 and variance 1: time step $1/n$, space step $1/\sqrt{n}$
 - Central limit theorem (CLT): $W_n(t) \rightarrow N(0, t)$ for each t ; Donsker: $W_n \Rightarrow W$ for the **whole path**, whatever the law of the steps Billingsley (1999)
- Right panel: $P(\max_{t \leq 1} W_n(t) > 1)$ rises from 0.220 ($n = 5$) to 0.316 ($n = 2000$), towards the reflection-principle value $2(1 - \Phi(1)) = 0.3173$ (Φ : standard Normal distribution function)
- A coarse grid misses the maximum between observations and convergence is slow, of order $1/\sqrt{n}$: daily closes understate the intraday range



Brownian Motion: Definition and Properties

- **Standard Brownian motion:** $W_0 = 0$; independent increments; $W_t - W_s \sim N(0, t - s)$ for $s < t$; continuous paths
 - ▶ Existence: [Wiener \(1923\)](#); simulation on a grid: $W_{t+\Delta t} = W_t + \sqrt{\Delta t} Z$, $Z \sim N(0, 1)$
- $\text{Cov}(W_s, W_t) = \min(s, t)$; a Gaussian process and a **martingale**: $\mathbb{E}[W_t | \mathcal{F}_s] = W_s$ (the efficient-market random walk of [Chapter 2](#))
- **Self-similarity:** $\{c^{-1/2} W_{ct}\}$ is again a standard Brownian motion for any $c > 0$
- Paths are continuous but **nowhere differentiable**
 - ▶ $(W_{t+h} - W_t)/h$ has standard deviation $1/\sqrt{h} \rightarrow \infty$: ordinary calculus fails

Brownian Paths



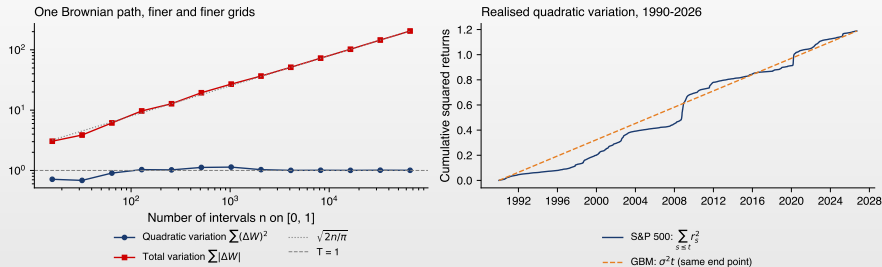
- Left: five paths with the pointwise band $\pm 1.96\sqrt{t}$ (95% for each t separately, not for the whole path); right: a path and its first 1% of time, stretched 100 times in time and 10 times in space
- The band widens like $\sqrt{t}$ (the square-root-of-time rule of [Chapter 7](#)) and a zoomed piece looks like the whole path; real markets break self-similarity, since fat tails and clustering depend on the horizon ([Chapter 1](#))



Total and Quadratic Variation

- Grid $0 = t_0 < t_1 < \dots < t_n = T$ with step $\Delta t = T/n$ and increments $\Delta W_k = W_{t_k} - W_{t_{k-1}}$
- **Total variation** $\sum_k |\Delta W_k|$: the length of the path
 - ▶ Its mean is $\sqrt{2nT/\pi} \rightarrow \infty$: a Brownian path has infinite length
- **Quadratic variation** $[W]_T = \lim \sum_k (\Delta W_k)^2 = T$
 - ▶ A random sum with a **deterministic** limit
 - ▶ $\text{Var}(Q_n) = 2T^2/n$: one year of daily data gives about 9% relative error ($\sqrt{2/252}$), the idea behind realised variance ([Chapter 9](#))
- Shorthand used in Itô calculus: $(dW_t)^2 = dt$

Quadratic Variation: Simulation and Data



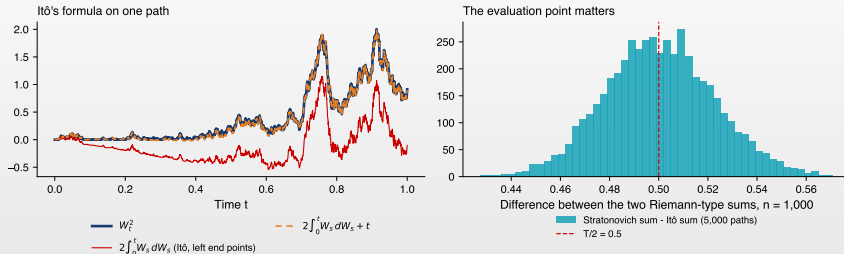
- Left: one Brownian path on grids with 16 to 65,536 intervals: the quadratic variation settles at 1.006 ($T = 1$), while the total variation keeps growing, 205 against $\sqrt{2n/\pi} = 204$
- Right: cumulative $\sum r_t^2$ of the S&P 500, 1.188 over 36.7 years (average volatility 18.0%); under geometric Brownian motion it would follow the straight line $\sigma^2 t$
- It rises in steps: September 2008 – March 2009 is 1.6% of the days but 15.2% of the quadratic variation; February – April 2020: 0.5% and 7.8%. Volatility is itself random (Heston, Section 10)



The Itô Integral

- Goal: give a meaning to $\int_0^T H_t dW_t$, the gain of holding H_t units of an asset whose price moves like W
 - ▶ Riemann–Stieltjes integration fails because the path has infinite total variation
- **Itô integral**: the limit in mean square of sums with the integrand at the **left** end point
 - ▶ $\int_0^T H_t dW_t = \lim_{n \rightarrow \infty} \sum_{k=1}^n H_{t_{k-1}} (W_{t_k} - W_{t_{k-1}})$, for H **adapted** with $\mathbb{E} \left[\int_0^T H_t^2 dt \right] < \infty$
 - ▶ Financial reading: H_t uses only information up to t ; the position is chosen before the next price move
- Zero mean, $\mathbb{E} \left[\int_0^T H_t dW_t \right] = 0$, and the **Itô isometry**, $\mathbb{E} \left[\left(\int_0^T H_t dW_t \right)^2 \right] = \mathbb{E} \left[\int_0^T H_t^2 dt \right]$
 - ▶ $t \mapsto \int_0^t H_s dW_s$ is a martingale: no strategy earns money from a martingale on average (weaker conditions:
▶ [Appendix](#))
- Mid-point sums give the **Stratonovich** integral, a different number; finance uses Itô, because a mid-point position would look into the future

Worked Example: $\int_0^T W_t dW_t$ on Simulated Paths



- Each term: $W_{t_{k-1}} \Delta W_k = \frac{1}{2}(W_{t_k}^2 - W_{t_{k-1}}^2) - \frac{1}{2}(\Delta W_k)^2$; the first part telescopes, the second tends to $\frac{1}{2}T$:

$$\int_0^T W_t dW_t = \frac{1}{2}W_T^2 - \frac{1}{2}T$$
 - Ordinary calculus would give $\frac{1}{2}W_T^2$: the extra $-\frac{1}{2}T$ is the quadratic variation, and the mean is $\frac{1}{2}(T - T) = 0$
- Left: on one path, $2\int_0^t W dW + t$ lies exactly on W_t^2 ; right: Stratonovich minus Itô sums over 5,000 paths, mean 0.500, s.d. 0.022 (theory $T/2 = 0.5$)
- Itô sums: mean 0.007, variance 0.520; theory: 0 and, by the isometry, $\int_0^1 t dt = 1/2$

Itô Processes and Itô's Lemma

- **Itô process:** $dX_t = a_t dt + b_t dW_t$ (drift a_t , diffusion coefficient b_t , both adapted)
 - ▶ Shorthand for $X_t = X_0 + \int_0^t a_s ds + \int_0^t b_s dW_s$
- **Itô's lemma:** for $f(t, x)$ twice differentiable [Itô \(1944\)](#)
 - ▶ $df(t, X_t) = (f_t + a_t f_x + \frac{1}{2} b_t^2 f_{xx}) dt + b_t f_x dW_t$
- Rule of thumb: Taylor expansion to second order with $dt \cdot dt = 0$, $dt \cdot dW = 0$, $dW \cdot dW = dt$
 - ▶ The second-order term $\frac{1}{2} b_t^2 f_{xx} dt$ does not vanish: this is the whole novelty

Quadratic Covariation and the Multivariate Itô Formula

- **Quadratic covariation** $[X, Y]_t = \lim \sum_k \Delta X_k \Delta Y_k$; if $\text{Corr}(dW^{(1)}, dW^{(2)}) = \rho$, then $d[W^{(1)}, W^{(2)}]_t = \rho dt$
 - ▶ Product rule: $d(X_t Y_t) = X_t dY_t + Y_t dX_t + d[X, Y]_t$
- **Multivariate Itô formula** for $X_t \in \mathbb{R}^d$ and $f(t, x)$ twice differentiable
 - ▶ $df(t, X_t) = f_t dt + \sum_i f_{x_i} dX_t^i + \frac{1}{2} \sum_{i,j} f_{x_i x_j} d[X^i, X^j]_t$
- Worked example (Heston, Section 10): $dS = \mu S dt + \sqrt{v} S dW^{(1)}$, $dv = \kappa(\theta - v) dt + \xi \sqrt{v} dW^{(2)}$
 - ▶ $d(S_t v_t) = S_t dv_t + v_t dS_t + \rho \xi v_t S_t dt$
 - ▶ ρ enters the drift of the product: with $\rho < 0$, price and variance tend to move in opposite directions

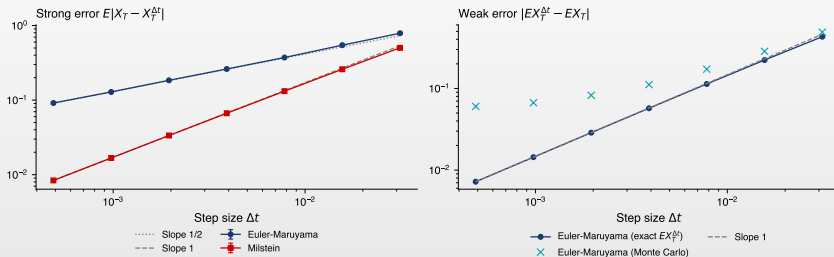
Stochastic Differential Equations and Two Schemes

- A **stochastic differential equation** (SDE): $dX_t = a(t, X_t) dt + b(t, X_t) dW_t$, X_0 given: drift and diffusion depend on the current state
 - ▶ Exact transitions for GBM, Ornstein–Uhlenbeck and Merton: simulate them exactly, with no discretisation error; Heston has only a closed-form characteristic function; existence conditions: [Appendix](#)
- **Euler–Maruyama**: freeze a and b over each step [Higham \(2001\)](#)
 - ▶ $X_{k+1} = X_k + a(X_k) \Delta t + b(X_k) \Delta W_k$, $\Delta W_k = \sqrt{\Delta t} Z_k$
- **Milstein**: add the next Itô–Taylor term [Milstein \(1975\)](#)
 - ▶ $X_{k+1} = X_k + a(X_k) \Delta t + b(X_k) \Delta W_k + \frac{1}{2} b(X_k) b'(X_k) [(\Delta W_k)^2 - \Delta t]$
 - ▶ The correction has conditional mean zero: it raises the strong order (path accuracy), not the weak order

Strong and Weak Convergence

- **Strong error:** $\mathbb{E}|X_T - X_T^{\Delta t}| \leq C \Delta t^\gamma$, how close each simulated path is to the true one
 - ▶ Euler–Maruyama $\gamma = 1/2$; Milstein $\gamma = 1$; needed for path-dependent quantities: hedging errors, barrier crossings
- **Weak error:** $|\mathbb{E}[g(X_T^{\Delta t})] - \mathbb{E}[g(X_T)]| \leq C \Delta t^\beta$, how close the distribution is
 - ▶ Both schemes: $\beta = 1$; enough for prices of European options and other expected values
 - ▶ Conditions: smooth coefficients with bounded derivatives and a smooth g Kloeden & Platen (1992); square-root coefficients (CIR, Heston), quantiles such as VaR and barrier crossings can converge more slowly
- Test equation Higham (2001): $dX = \lambda X dt + \mu X dW$, $\lambda = 2$, $\mu = 1$, $X_0 = 1$, $T = 1$, with the exact solution as benchmark
 - ▶ On a log-log plot, the order is the slope of the error against the step size (next slide)

Convergence of the Schemes

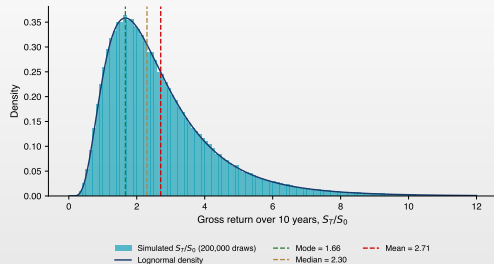
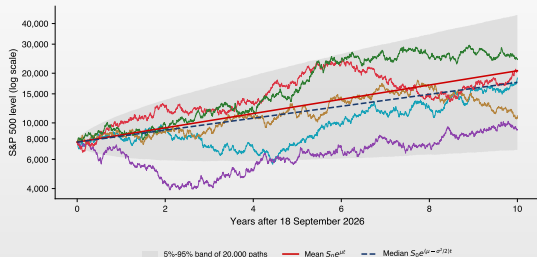


- 20,000 Brownian paths on a fine grid of 2^{11} steps, coarsened to steps from 2^{-11} to 2^{-5} ; strong orders (slope of the log-log fit, 95% CI by resampling whole paths): Euler–Maruyama 0.518 [0.512, 0.525], Milstein 0.986 [0.984, 0.987]
 - ▶ The narrow CIs measure Monte Carlo noise only; the small gap to 1/2 and 1 is finite-step curvature, not a different order; halving Δt halves the Milstein error, but cuts the Euler error only by 29%
- Weak order of Euler–Maruyama from the exact mean $(1 + \lambda \Delta t)^{T/\Delta t}$: 0.985, a deterministic slope
 - ▶ The Monte Carlo estimates flatten at small steps: their standard error, about 0.068, exceeds the bias; the sample size, not the step, is often the binding constraint [Glasserman \(2003\)](#)

Geometric Brownian Motion (GBM) from Itô's Lemma

- SDE: $dS_t = \mu S_t dt + \sigma S_t dW_t$: the **percentage** change has constant drift μ and volatility σ per year
 - ▶ Proposed by [Osborne \(1959\)](#) and [Samuelson \(1965\)](#); the price model behind [Black & Scholes \(1973\)](#)
- Itô's lemma for $f(x) = \ln x$, with $f_x = 1/x$, $f_{xx} = -1/x^2$, $a_t = \mu S_t$, $b_t = \sigma S_t$
 - ▶ $d \ln S_t = (\mu - \frac{1}{2}\sigma^2) dt + \sigma dW_t$, so $S_t = S_0 \exp((\mu - \frac{1}{2}\sigma^2)t + \sigma W_t)$
 - ▶ Prices stay positive; exact simulation needs only Normal draws
- The log price grows at the log drift $m = \mu - \sigma^2/2$, not at μ : volatility lowers the growth of the median investor's wealth
 - ▶ $\sigma = 18\%$ costs $\sigma^2/2 = 1.6$ percentage points a year; $\sigma = 67\%$ (Bitcoin), 22 points
- Testable implications: $r_t = \ln(S_t/S_{t-\Delta t})$ i.i.d. $N(m\Delta t, \sigma^2\Delta t)$ at every frequency
 - ▶ Skewness 0, excess kurtosis 0; no autocorrelation in r_t , $|r_t|$ or r_t^2 ; variance proportional to the horizon

Lognormal Outcomes: Ten Years of Simulated S&P 500 Paths



- GBM with $\hat{\mu} = 10.0\%$, $\hat{\sigma} = 18.0\%$ (1990–2026), $T = 10$ years: mean $\mathbb{E}[S_T] = S_0 e^{\mu T}$, median $S_0 e^{mT}$, mode $S_0 e^{(m-\sigma^2)T}$; gross returns 2.71, 2.30, 1.66 (right)
- Left: 20,000 exact paths from $S_0 = 7\,650$ (18 September 2026): mean 20 696, median 17 604, 5%–95% range 6 878 – 45 017; $P(S_T < S_0) = \Phi(-m\sqrt{T}/\sigma) = 7.1\%$
- $P(S_T < \mathbb{E}[S_T]) = \Phi(\sigma\sqrt{T}/2) = 61.2\%$: most investors end below the average, so report the median and quantiles; the fan is too narrow in crises (no fat tails, no clustering: tested below)

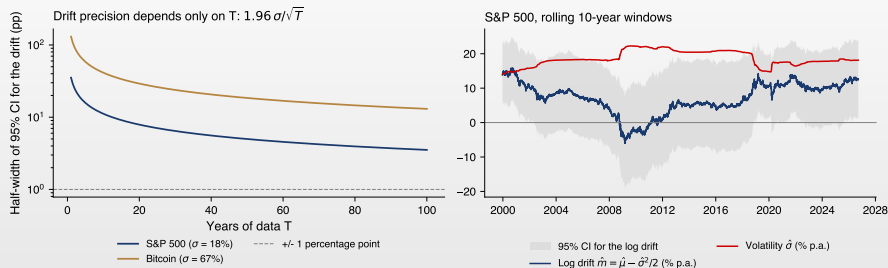
Estimating GBM by Maximum Likelihood

- Maximum likelihood estimation (MLE) from n log returns at step Δt , over $T = n\Delta t$ years: $\hat{\sigma}^2 = \frac{1}{n\Delta t} \sum_t (r_t - \bar{r})^2$, $\hat{m} = \bar{r}/\Delta t$, $\hat{\mu} = \hat{m} + \hat{\sigma}^2/2$
- Under GBM, the standard error (s.e.) $\text{s.e.}(\hat{\sigma}) = \sigma/\sqrt{2n}$ improves with more observations; $\text{s.e.}(\hat{m}) = \sigma/\sqrt{T}$ depends only on the **calendar span**, since $\sum_t r_t = \ln(S_T/S_0)$ uses only the first and the last price

	Years	$\hat{\sigma}$ (%)	$\text{s.e.}(\hat{\sigma})$	$\hat{m}$ (%)	$\text{s.e.}(\hat{m})$
S&P 500	36.7	18.0	0.13	8.3	3.0
BET	26.5	22.2	0.19	16.1	4.3
Bitcoin	12.0	66.6	0.71	43.1	19.2

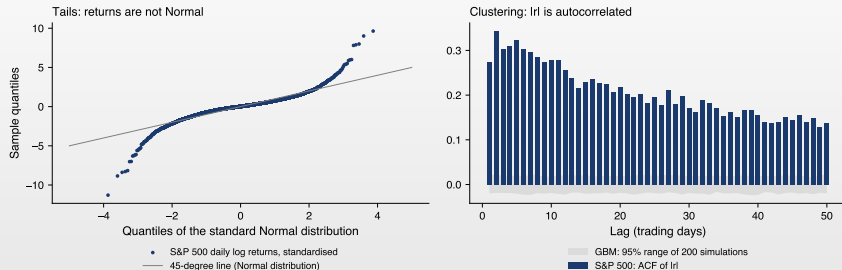
- Volatility is pinned down to 0.1–0.7 percentage points, the log drift only to 3–19 points; S&P 500: $\hat{\mu} = 8.3\% + 1.62\% = 10.0\%$ per year
- These s.e. are conditional on GBM: with excess kurtosis k , even independent returns inflate $\text{s.e.}(\hat{\sigma})$ by $\sqrt{(k+2)/2} = 2.5$ (S&P 500); block-bootstrap s.e. in Seminar 11, B2

How Precisely Can We Estimate the Drift?



- Left: half-width of the 95% CI for the drift against years of data; ± 1 point needs $(1.96 \hat{\sigma} / 0.01)^2 = 1243$ years for the S&P 500 and 17028 years for Bitcoin
- Right: S&P 500 on rolling 10-year windows: the log drift ranges from -6.0% (window ending 2009-03) to 15.8% (2000-09), each with a CI of about ± 11 points, so the variation is mostly noise; volatility stays between 14.1% and 22.3% : precise, but not constant
- Option pricing in [Chapter 12](#) avoids μ : under the risk-neutral measure only σ enters (Girsanov, Section 8)

Does GBM Fit the S&P 500?



- Left: quantile–quantile (QQ) plot of standardised daily log returns against the Normal distribution: excess kurtosis 10.9 instead of 0, worst day at -11.3 standard deviations, 62 days beyond ± 4 against 0.59 expected; Jarque–Bera (JB) statistic 45 805
- Right: autocorrelation function (ACF) of $|r_t|$: 0.27 at lag 1 and still 0.14 at lag 50, while 200 GBM samples of the same length never exceed 0.024 in absolute value
- GBM fails on both stylised facts of [Chapter 1 Cont \(2001\)](#); the rest of the chapter adds mean reversion, jumps and random volatility

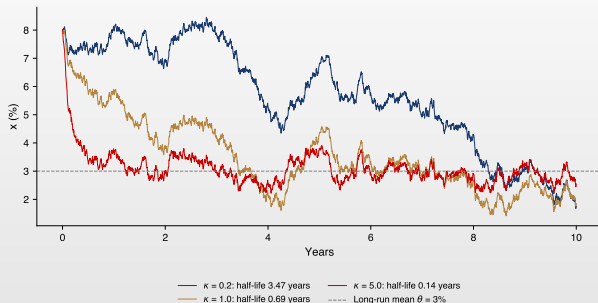
The Ornstein–Uhlenbeck (OU) Process

- SDE: $dx_t = \kappa(\theta - x_t) dt + \sigma dW_t$, $\kappa > 0$ [Uhlenbeck & Ornstein \(1930\)](#): long-run mean θ , speed of mean reversion κ , volatility σ
 - ▶ Above θ the drift is negative, below θ positive: a spring pulling x back; used for interest rates [Vasicek \(1977\)](#), volatility, spreads, commodity prices
- Itô's lemma on $e^{\kappa t} x_t$: $x_t = \theta + (x_0 - \theta)e^{-\kappa t} + \sigma \int_0^t e^{-\kappa(t-s)} dW_s$, a Normal variable with
 - ▶ $\mathbb{E}[x_t | x_0] = \theta + (x_0 - \theta)e^{-\kappa t}$, $\text{Var}(x_t | x_0) = \frac{\sigma^2}{2\kappa} (1 - e^{-2\kappa t})$
 - ▶ Stationary distribution ($t \rightarrow \infty$): $N(\theta, \sigma^2/(2\kappa))$
- **Half-life** $\ln 2/\kappa$: the time for an expected deviation from θ to halve
 - ▶ $\kappa = 0.5$ per year: half-life 1.39 years; unlike Brownian motion, the variance stops growing

Exact Discretisation: an AR(1)

- Sampled at step Δt the OU process is exactly an autoregressive model of order 1 (AR(1)):
 - ▶ $x_{t+\Delta t} = a + b x_t + \varepsilon_t$, $b = e^{-\kappa \Delta t}$, $a = \theta(1 - b)$, $\text{Var}(\varepsilon_t) = \frac{\sigma^2}{2\kappa}(1 - b^2)$
- Maximum likelihood = ordinary least squares (OLS) of $x_{t+\Delta t}$ on x_t , then invert the mapping
 - ▶ $\hat{\kappa} = -\ln \hat{b}/\Delta t$, $\hat{\theta} = \hat{a}/(1 - \hat{b})$, $\hat{\sigma} = \hat{s}_\varepsilon \sqrt{2\hat{\kappa}/(1 - \hat{b}^2)}$
- Standard errors for $\hat{\kappa}$ and $\hat{\theta}$ by the delta method from the OLS covariance of $(\hat{a}, \hat{b})$ (Seminar 11, A9)
- With daily data b is very close to 1: κ is hard to estimate, as with unit-root tests

Three Speeds of Mean Reversion

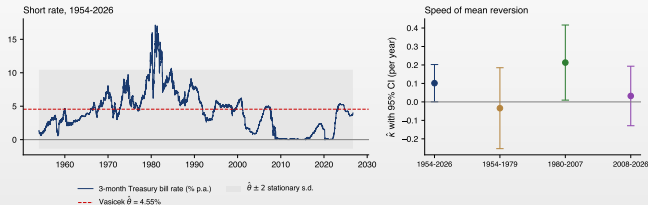


- OU paths driven by the same shocks, starting at 8%, with $\theta = 3\%$, $\sigma = 1\%$ and $\kappa = 0.2, 1, 5$; exact simulation, daily steps
- Half-lives 3.47, 0.69 and 0.14 years: with $\kappa = 5$ the gap from 8% to 3% closes within a few months
- Stationary standard deviations $\sigma / \sqrt{2\kappa}$: 1.58%, 0.71%, 0.32%; slow reversion looks like a random walk for years
- Long-horizon forecasts converge to θ

Short-Rate Models: Vasicek and CIR

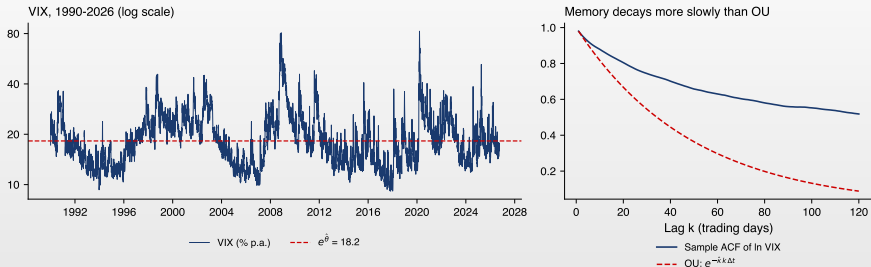
- **Vasicek**: the short rate r_t is an OU process [Vasicek \(1977\)](#)
 - ▶ Normal distribution: negative rates have positive probability $\Phi(-\theta\sqrt{2\kappa}/\sigma)$ in the long run
 - ▶ Bond prices and yields in closed form (Feynman–Kac, Section 8)
- **Cox–Ingersoll–Ross (CIR)**: $dr_t = \kappa(\theta - r_t)dt + \sigma\sqrt{r_t}dW_t$ [Cox, Ingersoll & Ross \(1985\)](#)
 - ▶ Volatility shrinks near zero; the rate stays positive if $2\kappa\theta \geq \sigma^2$ (Feller condition) [Feller \(1951\)](#)
- Which is closer to the data is an empirical question, tested in Section 7 [Chan, Karolyi, Longstaff & Sanders \(1992\)](#)

Vasicek on the 3-Month Treasury Bill Rate



- Left: daily 3-month US Treasury bill rate (DTB3), 1954–2026, with the Vasicek long-run mean $\hat{\theta} = 4.55\%$; $\hat{\kappa} = 0.101$ (s.e. 0.051), half-life 6.8 years, $\hat{\sigma} = 1.33\%$, daily $\hat{b} = 0.99960$: almost a unit root
- Right: $\hat{\kappa}$ with 95% CI by period: -0.034 (1954–1979, explosive), 0.213 (1980–2007), 0.032 (2008–2026); two of the three CIs include zero
- In finite samples $\hat{\kappa}$ overstates κ by roughly $4/T$ (T in years) [Tang & Chen \(2009\)](#); with the full-sample estimates as the truth, 500 samples of 72 years give a mean $\hat{\kappa} = 0.164$: bias 0.063 against $4/T = 0.055$, with 78% of the estimates above the truth
 - The bias depends on the calendar span, not on the sampling frequency: estimated half-lives are too short and mean reversion looks faster than it is (Seminar 11, B4)

Is Volatility Mean-Reverting? The VIX



- Left: VIX 1990–2026 on a log scale with the OU long-run level $e^{\hat{\theta}} = 18.2$; OU fit to $\ln VIX$: $\hat{\kappa} = 5.10$ (s.e. 0.53), half-life 34 trading days: spikes such as 82.7 (2020-03-16) fade within weeks
- Right: sample ACF of $\ln VIX$ against the OU-implied $e^{-\hat{\kappa}k\Delta t}$: at lag 60, 0.63 in the data against 0.30 under OU: a fast decay, then a slow component
- One OU factor is too simple: candidates are two-factor or long-memory models, while rough volatility (Section 12) is a distinct alternative; mean-reverting volatility is the building block of Heston

Infill versus Long-Span Asymptotics

Infill: $\Delta \rightarrow 0$ on a fixed $[0, T]$

- ▣ $\sum_k (\Delta X_k)^2 \rightarrow \int_0^T b_t^2 dt$
- ▣ The diffusion coefficient is identified without error in the limit: quadratic variation is observable (continuous Itô process, no measurement noise)
- ▣ Sampling more often helps σ (Section 5) and $b(x)$ (nonparametric, below)

Long span: $T \rightarrow \infty$

- ▣ The drift is not consistently estimable on a fixed span: its Fisher information grows with T , not with n
- ▣ GBM: $\text{s.e.}(\hat{m}) = \sigma/\sqrt{T}$; OU: $\text{Var}(\hat{\kappa}) \approx (e^{2\kappa\Delta} - 1)/(T\Delta) \rightarrow 2\kappa/T$ as $\Delta \rightarrow 0$ (Seminar 11, A9)
- ▣ The bias of $\hat{\kappa}$ does not shrink with finer sampling Yu (2012); $b = e^{-\kappa\Delta} \rightarrow 1$, so $\hat{\kappa}$ is far from Normal when κT is small

- ▣ One phenomenon, not two facts: the imprecise GBM drift and the biased $\hat{\kappa}$ are both long-span problems

Likelihood for Nonlinear Diffusions

- **Euler pseudo-likelihood:** a Gaussian transition $r_{t+\Delta} | r_t \sim N(r_t + a(r_t)\Delta, b^2(r_t)\Delta)$
 - ▶ Biased at order Δ : small with daily data, larger with monthly or quarterly steps
- **Exact CIR:** $2c r_{t+\Delta} | r_t$ is noncentral χ^2 with $4\kappa\theta/\sigma^2$ degrees of freedom [Cox, Ingersoll & Ross \(1985\)](#)
 - ▶ $c = 2\kappa/(\sigma^2(1 - e^{-\kappa\Delta}))$, noncentrality $2c r_t e^{-\kappa\Delta}$; Feller condition $\Leftrightarrow$ degrees of freedom ≥ 2 (Seminar 11, A5)
- General diffusions: closed-form Hermite expansions of the transition density [Aït-Sahalia \(2002\)](#)
- CKLS (Chan–Karolyi–Longstaff–Sanders): $dr = \kappa(\theta - r)dt + \sigma r^\gamma dW$ nests Vasicek ($\gamma = 0$) and CIR ($\gamma = 1/2$) [Chan, Karolyi, Longstaff & Sanders \(1992\)](#)
 - ▶ Gaussian (Euler) quasi-likelihood; tests on γ with sandwich standard errors whose score covariance is heteroskedasticity- and autocorrelation-consistent (HAC) [Newey & West \(1987\)](#)

Vasicek, CIR and CKLS on the T-Bill Rate, 1954–2007

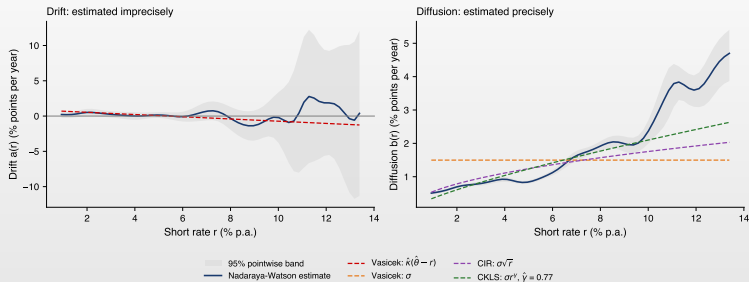
Model	$\hat{\kappa}$ daily (s.e.)	$\hat{\kappa}$ monthly (s.e.)	γ
Vasicek, exact	0.158 (0.073)	0.201 (0.083)	0
Vasicek, Euler	0.158 (0.073)	0.199 (0.082)	0
CIR, exact	0.113 (0.060)	0.153 (0.070)	1/2
CIR, Euler	0.093 (0.060)	0.144 (0.069)	1/2
CKLS, Euler	0.080 (0.057)	0.144 (0.068)	0.77 (0.033); 0.60 (0.149)

- Daily DTB3 ($n = 13\,487$) and month-end values ($n = 647$); 2008–2026 excluded: exact zeros and negative quotes lie outside the domain of these likelihoods (▶ [Appendix](#))
- Exact and Euler differ by less than one standard error: discretisation error is small next to the long-span uncertainty of $\hat{\kappa}$
- Volatility rises with the level: HAC Wald tests reject $\gamma = 0$ at both frequencies and $\gamma = 1/2$ with daily but not with monthly data (statistics: ▶ [Appendix](#))

Nonparametric Drift and Diffusion

- No functional form: kernel averages of the increments around each level x
 - ▶ $\hat{a}(x) = \sum_t w_t(x) \Delta r_t / \Delta$, $\hat{b}^2(x) = \sum_t w_t(x) (\Delta r_t)^2 / \Delta$, $w_t(x) = K((r_t - x)/h) / \sum_s K((r_s - x)/h)$
 - ▶ Nadaraya–Watson weights summing to one, Gaussian kernel K , bandwidth h [Stanton \(1997\)](#)
- $\mathbb{E}[\Delta r \mid r = x] / \Delta = a(x) + O(\Delta)$ and $\mathbb{E}[(\Delta r)^2 \mid r = x] / \Delta = b^2(x) + O(\Delta)$: exact only as $\Delta \rightarrow 0$
 - ▶ The diffusion converges at the infill rate, the drift only at the long-span rate; valid for non-stationary but recurrent processes [Bandi & Phillips \(2003\)](#)
- Caution: nonparametric specification tests reject far too often with persistent data [Pritsker \(1998\)](#)

Nonparametric Drift and Diffusion of the T-Bill Rate



- Daily DTB3, 1954–2007; bandwidth $h = 0.44$ percentage points; 95% heteroskedasticity-robust pointwise bands (serial dependence ignored)
- Drift: zero lies inside the band at 97% of the levels; diffusion: median relative s.e. 8%, $b(r)$ rises from 0.5% to 4.7% (log-slope 0.90)
- No parametric shape fits everywhere: inside the band at 35% (CKLS), 7% (CIR), 5% (Vasicek) of the levels, a descriptive share

Girsanov's Theorem

- Real-world measure P and an equivalent measure Q (the same events of probability zero), linked by a **Radon–Nikodym density**
 - ▶ $\frac{dQ}{dP} \Big|_{\mathcal{F}_T} = Z_T = \exp \left(- \int_0^T \theta_s dW_s - \frac{1}{2} \int_0^T \theta_s^2 ds \right)$
- **Girsanov (1960)**: under Q , $\tilde{W}_t = W_t + \int_0^t \theta_s ds$ is a Brownian motion if $\mathbb{E} \left[e^{\frac{1}{2} \int_0^T \theta_s^2 ds} \right] < \infty$ (Novikov condition)
- For $dX = a dt + b dW$: $dX = (a - b\theta) dt + b d\tilde{W}$
 - ▶ **Only the drift changes**: the diffusion coefficient and the quadratic variation are the same under P and Q
 - ▶ Under GBM (constant σ), realised variance ([Chapter 9](#)) measures the σ that prices options, while μ drops out; with stochastic volatility, P and Q weigh future variance paths differently, so historical variance is a benchmark, not a calibration

Market Price of Risk: P versus Q

- GBM: $\theta = (\mu - r)/\sigma$ is the **market price of risk**; under Q , $dS = rS dt + \sigma S d\tilde{W}$ and $e^{-rt}S_t$ is a Q -martingale (Seminar 11, A1)
 - ▶ S&P 500, 1990–2026: $\hat{\theta} = (10.0\% - 2.7\%)/18.0\% = 0.40$, s.e. $1/\sqrt{T} = 0.17$ (r : mean 3-month T-bill rate): as imprecise as the drift
- Vasicek with risk premium $\lambda\sigma$: $\kappa^Q = \kappa$, $\kappa\theta^Q = \kappa\theta - \lambda\sigma$
 - ▶ The long-run level that prices bonds differs from the one estimated on data (Section 6)
- Heston with volatility risk premium λv_t : $\kappa^Q = \kappa + \lambda$, $\kappa^Q\theta^Q = \kappa\theta$
 - ▶ $\lambda < 0$ (investors pay for protection against variance) gives $\theta^Q > \theta$: the variance risk premium of Chapter 9

Feynman–Kac: From Expectations to the Pricing PDE

- If $dX = a(X) dt + b(X) d\tilde{W}$ under Q Kac (1949)
 - ▶ $u(t, x) = \mathbb{E}^Q \left[e^{-\int_t^T r_s ds} g(X_T) \mid X_t = x \right]$ solves $u_t + a u_x + \frac{1}{2} b^2 u_{xx} - r u = 0$, $u(T, x) = g(x)$
- Proof idea: $e^{-\int_0^t r_s ds} u(t, X_t)$ is a Q -martingale, so Itô's lemma must give it a zero drift
- Vasicek bond ($g = 1$, $X = r$): $p(t, r; T) = \exp(A(\tau) - B(\tau)r)$, $\tau = T - t$ (Seminar 11, A3)
 - ▶ $B(\tau) = \frac{1 - e^{-\kappa\tau}}{\kappa}$, $A(\tau) = \left(\theta^Q - \frac{\sigma^2}{2\kappa^2} \right) (B(\tau) - \tau) - \frac{\sigma^2 B(\tau)^2}{4\kappa}$
 - ▶ Yields $y(\tau) = (B(\tau)r - A(\tau))/\tau$ tend to $\theta^Q - \sigma^2/(2\kappa^2)$: rate uncertainty lowers long yields (convexity); comparison with Treasury yields: ▶ Appendix
- Chapter 12 derives Black–Scholes by the same route

Merton's Jump-Diffusion

- Brownian paths are continuous, but prices gap on news: on 19 October 1987 the Dow Jones fell 22.6%, about 20 standard deviations under GBM with 18% volatility
 - ▶ A jump cannot be hedged by continuous trading [Merton \(1976\)](#)
- Log price = Brownian motion with drift + **compound Poisson** jumps: $d \ln S_t = m dt + \sigma dW_t + dJ_t$,
 $J_t = \sum_{i \leq N_t} Y_i$
 - ▶ N_t Poisson with intensity λ ; sizes $Y_i \sim N(\mu_J, \sigma_J^2)$ i.i.d.; W , N and Y independent; five parameters $(m, \sigma, \lambda, \mu_J, \sigma_J)$
- Return over Δt given k jumps: $N(m\Delta t + k\mu_J, \sigma^2\Delta t + k\sigma_J^2)$; unconditionally a **Poisson mixture** of Normal densities, i.i.d. over time
 - ▶ $f(r) = \sum_{k \geq 0} e^{-\lambda\Delta t} \frac{(\lambda\Delta t)^k}{k!} \varphi(r; m\Delta t + k\mu_J, \sigma^2\Delta t + k\sigma_J^2)$
- Exact simulation: draw $N \sim \text{Poisson}(\lambda\Delta t)$, then draw from the Normal distribution with the conditional moments

Moments of the Merton Return

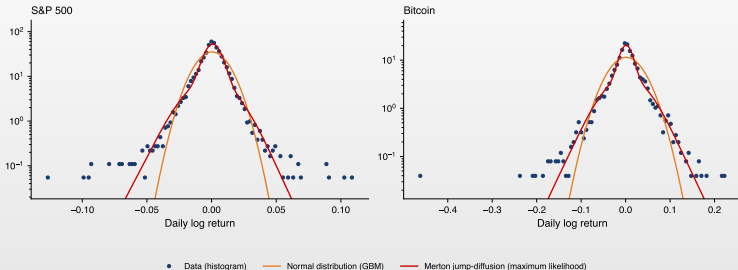
- Cumulants of the return over Δt (the k -th cumulant of a compound Poisson sum is $\lambda \Delta t \mathbb{E}[Y^k]$):
 - ▶ $\kappa_2 = \sigma^2 \Delta t + \lambda \Delta t (\mu_J^2 + \sigma_J^2)$, $\kappa_3 = \lambda \Delta t (\mu_J^3 + 3\mu_J \sigma_J^2)$
 - ▶ $\kappa_4 = \lambda \Delta t (\mu_J^4 + 6\mu_J^2 \sigma_J^2 + 3\sigma_J^4)$; skewness $\kappa_3 / \kappa_2^{3/2}$, excess kurtosis κ_4 / κ_2^2
- Negative μ_J gives negative skewness; any jump gives positive excess kurtosis
- Excess kurtosis falls like $1/\Delta t$ as the horizon grows: jumps matter most at short horizons
 - ▶ Example: $\sigma = 15\%$, $\lambda = 5$, $\mu_J = -3\%$, $\sigma_J = 4\%$: daily excess kurtosis 17.6, skewness -2.07 (Seminar 11, A7)

Maximum Likelihood Estimates

	$\hat{\sigma}$ (%)	$\hat{\lambda}$ (per year)	$\hat{\mu}_J$ (%)	$\hat{\sigma}_J$ (%)	Jump share of variance	LR
S&P 500	9.4	102 (11)	-0.16	1.44	71%	2 401
BET	11.3	79 (7)	-0.09	2.07	73%	2 326
Bitcoin	24.6	239 (21)	0.00	3.85	85%	1 357

- Log-likelihood maximised numerically over the Poisson mixture (up to 12 jumps per day); standard errors from the Hessian in brackets
- The likelihood is **unbounded**: with $m\Delta t$ equal to one observed return and $\sigma \rightarrow 0$ that density explodes, so the MLE is a local maximum [Honoré \(1998\)](#)
 - ▶ The LR test against GBM ($\lambda = 0$) is defined on $\sigma \geq 5\%$ and is not χ^2_3 : $\lambda = 0$ lies on the boundary and μ_J, σ_J are unidentified; bootstrap critical values in Seminar 11, B6
- The fit prefers **many small** jumps (about 80–240 per year), not a few crashes

Merton Fit to the Return Distribution

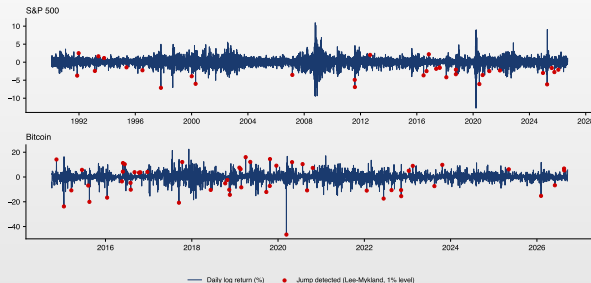


- Density of daily log returns on a log scale: histogram, the Normal distribution implied by GBM and the fitted Merton mixture; Merton follows the peak and the shoulders far better, and the LR statistics in the thousands reject GBM
- S&P 500: implied excess kurtosis 3.7 against 10.9 in the data, skewness -0.31 against -0.36 ; the most extreme days lie beyond both fitted curves
- About 102 “jumps” a year: the model uses jumps to mimic **changing volatility**; an i.i.d. model cannot tell a jump from a volatile period, so a direct test is needed

Detecting Jumps: the Lee–Mykland Test

- Idea: a jump is a return that is huge **relative to the local volatility** Lee & Mykland (2008)
 - ▶ $L_t = r_t / \hat{\sigma}_t$, $\hat{\sigma}_t^2 = \frac{1}{K-2} \sum_{j=t-K+2}^{t-1} |r_j| |r_{j-1}|$ (bipower variation, robust to jumps)
- Under no jumps, the maximum of $|L_t|$ over n days has a Gumbel limit
 - ▶ Reject if $(|L_t| - C_n) / S_n > -\ln(-\ln(1 - \alpha))$; at $\alpha = 1\%$ this threshold for the **normalised** statistic, not for $|L_t|$, is 4.60
 - ▶ $C_n = \frac{\sqrt{2 \ln n}}{c} - \frac{\ln \pi + \ln \ln n}{2c\sqrt{2 \ln n}}$, $S_n = \frac{1}{c\sqrt{2 \ln n}}$, $c = \sqrt{2/\pi}$ (the mean of $|Z|$, which rescales the bipower estimate), n = number of tested days
- Daily data: window $K = 16$ days, as recommended by the authors
- Controls, asymptotically, the probability of a false jump over the **whole** sample, not day by day

Jumps Detected in the S&P 500 and Bitcoin



	S&P 500	Bitcoin
Candidates	32	52
Per year	0.87	4.35
Negative (%)	84	52
In the top 10 moves	0	5

- Daily log returns with the days flagged at the 1% level
- S&P 500: the largest moves came in turbulent periods (2008, 2020) and are not flagged, while a 1.2% move in a calm market is (1993-10-14)
- Candidate detections are not the fitted Merton intensity: the gap may reflect stochastic volatility, drift bursts (short explosive trends in the drift) and limited detection power in daily data

 MFM_ch11_jumps

The Heston Model

- Two SDEs, for the log price and for the instantaneous variance v_t [Heston \(1993\)](#)
 - ▶ $d \ln S_t = (\mu - \frac{1}{2} v_t) dt + \sqrt{v_t} dW_t^{(1)}$
 - ▶ $dv_t = \kappa(\theta - v_t) dt + \xi \sqrt{v_t} dW_t^{(2)}, \text{Corr}(dW^{(1)}, dW^{(2)}) = \rho$
- Variance follows a CIR process: mean-reverting to θ with speed κ ; ξ = volatility of volatility
 - ▶ Feller condition $2\kappa\theta \geq \xi^2$: variance never touches zero [Feller \(1951\)](#)
- $\rho < 0$: **leverage effect**, falling prices come with rising volatility
 - ▶ Simulation: Euler scheme with full truncation $v^+ = \max(v, 0)$ [Lord, Koekkoek & van Dijk \(2010\)](#); an alternative discretisation, the quadratic-exponential scheme [Andersen \(2008\)](#)

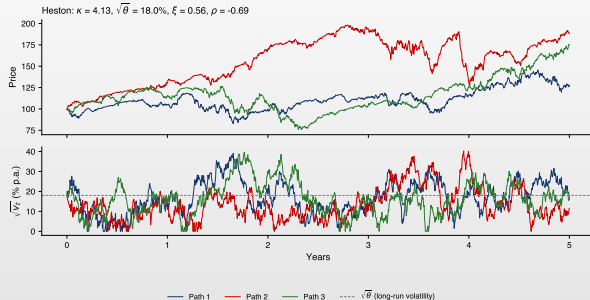
Heston Parameters from the VIX

- Proxy $y_t = (\text{VIX}_t/100)^2$ in place of v_t ; discretised CIR regression without an extra intercept
 - ▶ $\frac{y_{t+\Delta t} - y_t}{\sqrt{y_t}} = \kappa\theta \frac{\Delta t}{\sqrt{y_t}} - \kappa\Delta t\sqrt{y_t} + \xi\sqrt{\Delta t}\varepsilon_t$; ρ = correlation of return and variance shocks
- S&P 500 and VIX, 1990–2026: $\hat{\kappa} = 4.13$ (half-life 42 days), $\hat{\xi} = 0.56$, $\hat{\rho} = -0.69$
 - ▶ $\sqrt{\hat{\theta}} = 20.9\%$ from the VIX, but the realised volatility is 18.0%; Feller ratio of the proxy $2\hat{\kappa}\hat{\theta}/\hat{\xi}^2 = 1.16$
- But y_t is **not** v_t : it is a risk-neutral expectation of future variance, affine in v_t (next slide)
 - ▶ For real-world paths we set θ to the realised variance; $\hat{\xi}$ is attenuated relative to ξ

What the VIX Identifies in Heston

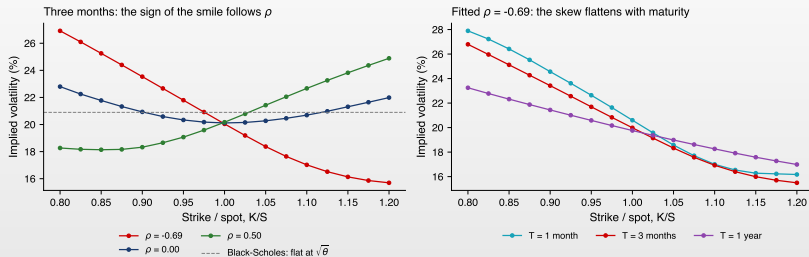
- Under Q : $y_t = \frac{1}{\tau} \mathbb{E}^Q \left[\int_t^{t+\tau} v_s ds \mid \mathcal{F}_t \right] = a + b v_t$, $\tau = 30/365$ (Seminar 11, A6)
 - ▶ $b = \frac{1 - e^{-\kappa^Q \tau}}{\kappa^Q \tau} < 1$, $a = \theta^Q(1 - b)$: **affine** in v_t , with Q -parameters
- What the regression on y_t recovers
 - ▶ κ : the P -speed, since an affine map keeps the dynamics of v_t ; ρ : unchanged, since correlation is scale-free, but estimating it needs paired return and variance shocks, and daily proxy changes only approximate them
 - ▶ ξ : attenuated; the diffusion of y is $\xi \sqrt{b(y - a)}$, so $\xi_y = b \xi \sqrt{v/(a + bv)}$ varies with the state and equals $b \xi$ only at $v_t = \theta^Q$ (local correction $\hat{\xi}/b$: ▶ Appendix)
 - ▶ The long-run level $a + b \theta^P$ mixes θ^Q and θ^P : recover $\theta^P = (\theta_y - a)/b$ before a Feller check on v_t
- Proper estimation fits the shifted square-root dynamics of y , or v_t and the observation equation jointly
- κ^Q is not identified from the VIX alone: joint P/Q estimation with option prices [Chernov & Ghysels \(2000\)](#), or moments of realised variance [Bollerslev & Zhou \(2002\)](#)

Simulated Heston Paths



- Three five-year paths with the VIX-based $\hat{\kappa}$, $\hat{\xi}$ (uncorrected), $\hat{\rho}$ and θ equal to the realised S&P 500 variance; top: price, bottom: $\sqrt{v_t}$
- Clustering now comes from the model; price falls coincide with volatility spikes (simulated return–variance correlation -0.69 , the leverage effect through ρ)
- Feller ratio $2\hat{\kappa}\theta/\hat{\xi}^2 = 0.86 < 1$: the CIR variance can reach zero; negative values arise only in the Euler approximation, and full truncation sets them to zero (1.4% of the steps)

A Preview of Chapter 12: the Volatility Smile



- Implied volatility: the σ that makes the Black–Scholes price equal to a given option price, flat across strikes under GBM; here from Heston call prices (Monte Carlo, 100,000 antithetic paths)
- Left, three months: with $\hat{\rho} = -0.69$, 26.9% at strike 80% of spot, 20.0% at the money, 15.7% at 120%: a downward **skew**; with $\rho = 0$ a symmetric **smile** (22.8%, 20.1%, 22.0%)
- Right: the skew flattens with maturity, from 27.9% – 16.2% at one month to 23.2% – 17.0% at one year; illustration only: pricing needs κ^Q , θ^Q calibrated to option prices ([Chapter 12](#))

Affine Models and the GARCH Diffusion Limit

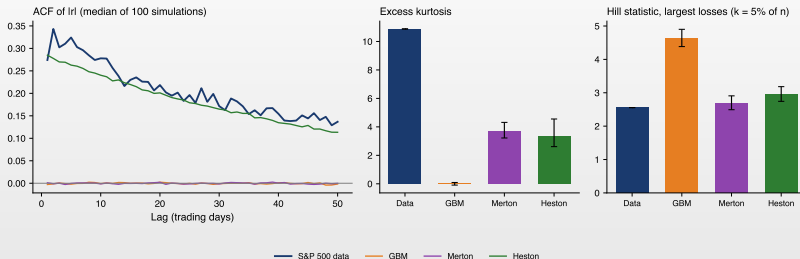
Affine jump-diffusions Duffie, Pan & Singleton (2000)

- Drift, variance and jump intensity affine in the state X_t : Vasicek, CIR, Merton, Heston and Bates
- $\mathbb{E} \left[e^{iu^\top X_\tau} \mid X_t \right] = \exp \left(\alpha(\tau) + \beta(\tau)^\top X_t \right)$, with α, β solving Riccati ordinary differential equations
- Option prices by Fourier inversion; the Vasicek $A(\tau)$, $B(\tau)$ of Section 8 are the simplest case

From GARCH to continuous time Nelson (1990)

- As the step shrinks, GARCH(1,1) converges to $dv_t = \kappa(\theta - v_t) dt + \xi v_t dW_t^{(2)}$: like Heston, with v_t instead of $\sqrt{v_t}$
- $\kappa \approx (1 - \alpha - \beta)/\Delta t$: $\alpha + \beta = 0.99$ with daily data gives $\kappa \approx 2.5$ per year, a half-life of about 70 days (Chapter 5)
- GARCH and its limit are not asymptotically equivalent experiments: inference can differ even as $\Delta t \rightarrow 0$ Wang (2002)

Which Model Reproduces the S&P 500?



- 100 simulated samples of 9 245 days from each fitted model: median ACF of $|r_t|$, excess kurtosis and the Hill statistic of the largest losses ($k = 5\%$ of the returns; lower = heavier tail), with 90% ranges; data: kurtosis 10.88, Hill 2.56
- Merton gets the Hill statistic (2.70) and skewness (-0.32 against -0.36) but no clustering; Heston, with the VIX-based parameters, gets the clustering but no skewness (0.00)
- No fitted model reaches the kurtosis of the data; the evidence of Andersen, Benzoni & Lund (2002) and Eraker, Johannes & Polson (2003) favours jumps and stochastic volatility (SVJ, Bates (1996)); full table in the [Appendix](#)

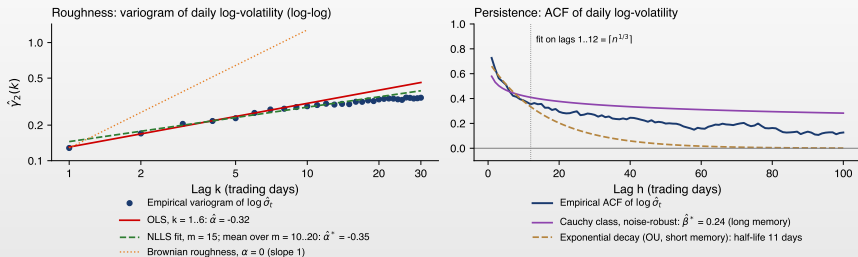
Which Model When? Features, Identification and Limits

Feature in the data	Model	What the data identify	Unresolved
Positive prices, $\sqrt{t}$ scaling	GBM	σ precisely; μ only over very long spans	fat tails, clustering
Mean reversion of rates and of the VIX	OU, Vasicek, CIR, CKLS	$b(r)$ precisely; κ biased upwards, needs a long span	one factor too simple for the VIX
Gaps, short-horizon fat tails	Merton	λ, σ_J only under i.i.d. returns; the jump share depends on the frequency	no clustering
Clustering, leverage, implied-volatility skew	Heston	κ, ρ from the VIX; ξ attenuated; $\theta^Q \neq \theta^P$	kurtosis and skewness of the data
All of the above	Bates (Heston + jumps)	returns and option surfaces jointly	latent states: filtering needed

- Roughness and long memory of volatility, next, are features none of these one-factor models produces

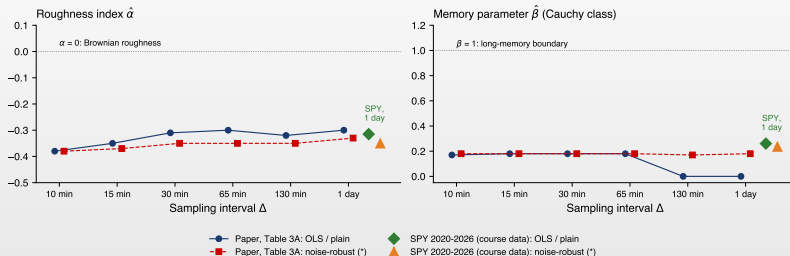
Research Question: Rough, Yet Persistent?

- Paper: [Bennedsen, Lunde & Pakkanen \(2022\)](#), *Journal of Financial Econometrics*
 - ▶ It builds on [Gatheral, Jaisson & Rosenbaum \(2018\)](#): log-volatility as a fractional OU process with Hurst exponent $H < 1/2$ (RFSV), whereas the long-memory model of [Comte & Renault \(1998\)](#) uses $H > 1/2$ (Introduction)
- Roughness index α : $1 - \rho(h) \sim c|h|^{2\alpha+1}$ as $h \rightarrow 0$, with ρ the autocorrelation of log-volatility (eq. (1.1)); $\alpha < 0$: paths rougher than Brownian motion
 - ▶ Memory parameter β : $\rho(h) \sim c|h|^{-\beta}$ as $h \rightarrow \infty$ (eq. (2.2)); $\beta < 1$: the ACF is not integrable, i.e. long memory
- The problem: in the fractional OU process a single H fixes both, $\alpha = H - 1/2$ and $\beta = 2 - 2H$ (Section 1): with $H < 1/2$, $\beta > 1$, so a rough fractional OU process cannot have long memory
 - ▶ Research question: are realised volatilities both rough and persistent, and do models that set α and β separately forecast better? (Introduction, Sections 4 and 5)
- Here: the daily part of the analysis on SPY; the model, the estimators and the forecast results of the paper are in the [Appendix](#), the intraday extension in Seminar 11, C3

Case Study: Replication on SPY, $\Delta = 1$ Day

- SPY 5-minute bars, 1 468 full days (October 2020 – September 2026); $x_t = \log \hat{\sigma}_t = \frac{1}{2} \log BV_t$, daily bipower variation of the 78 five-minute returns [Barndorff-Nielsen & Shephard \(2004\)](#)
- Left, rough: variogram $\hat{\gamma}_2(k) = \frac{1}{n-k} \sum_t (x_{t+k} - x_t)^2$; $\hat{\alpha}_{OLS} = -0.32$ ($k = 1, \dots, 6$), noise-robust $\hat{\alpha}_{NLLS}^* = -0.35$; the paper at 1 day: $-0.30, -0.33$ (Table 3, Panel A)
- Right, persistent: ACF 0.13 at lag 100; Cauchy $\hat{\beta} = 0.26$, $\hat{\beta}^* = 0.24$ on 12 lags, below 1: consistent with long memory, not a proof; an OU fit (half-life 11 days) implies 0.001 at lag 100

Case Study: SPY Against Table 3 of the Paper



- Lines: E-mini S&P 500 futures, 2011–2014, Δ from 10 minutes to 1 day (Bennedsen, Lunde & Pakkanen (2022), Table 3, Panel A); markers: SPY at 1 day, 2020–2026, course data; * = noise-robust
- $\hat{\alpha}$ barely moves with Δ (−0.38 to −0.30): roughness is a property of the process, and SPY falls in the same band; $\hat{\beta}^* = 0.17$ –0.18 at every Δ , far below 1, while the plain $\hat{\beta}$ is unstable
- Limits: another contract, period and estimator (bars, no pre-averaging); realised measures estimate integrated, not spot, variance

An Open Question: How Much of the Price Path Is Discontinuous?

- Model: $dX_t = \mu_t dt + \sigma_t dW_t + J_t dN_t$; the answer depends on the sampling frequency and on the model
 - ▶ Daily Merton MLE (this chapter): jumps carry 71% of S&P 500 variance; the 5-minute bipower test (Chapter 9): 2.8% for SPY
 - ▶ Christensen, Oomen & Podolskij (2014): at tick frequency most jumps detected at coarser sampling disappear; they are rapid but continuous moves
 - ▶ Christensen, Oomen & Renò (2022): **drift bursts**, short explosive trends in μ_t , look like jumps in coarse data
- Formal question: $H_0: \lambda = 0$ (continuous path with stochastic volatility) against $H_1: \lambda > 0$
 - ▶ Non-standard: $\lambda = 0$ lies on the boundary and (μ_J, σ_J) are unidentified under H_0 , so the LR statistic is not χ^2

The Discovery Loop and What the Researcher Checks

The loop with an AI assistant

- Assistant: an LLM such as Claude, ChatGPT or Gemini; **1. Literature:** tabulate published jump shares by frequency; check every DOI on [Crossref](#)
- **2. Hypotheses:** Merton jumps, Heston volatility, drift bursts, each with a distinct testable implication
- **3. Replicate first:** $\hat{\lambda} = 102$ per year, jump share 71%
- **4. Pre-registered robustness, 5. AI as a hostile referee, 6. Report** every specification, the code and the AI log (prompts: [Appendix](#))

Checks: failure and test

- Boundary, unidentified parameters: parametric bootstrap under H_0
- Look-ahead in the volatility filter: standardise only with $\mathcal{F}_{t-1}$
- Frequency mismatch: jump shares at every frequency on a common sample
- Multimodal likelihood: many starting points and a profile in λ
- Invented numbers: every claim needs a table and a DOI

Mini-Case: Jumps or Stochastic Volatility in Disguise?

- Replication: Merton on S&P 500 daily returns, $\hat{\lambda} = 102$ per year, jump share 71% (Section 9)
- A typical assistant proposal: “71% of variance comes from jumps”. Rival hypothesis: the Poisson mixture absorbs time-varying volatility; then standardising by EWMA volatility ($\lambda_{\text{EWMA}} = 0.94$, known at $t - 1$) removes the jumps

	Raw returns	EWMA-standardised
Excess kurtosis	10.9	2.9
LR, Merton vs the Normal distribution	2 401	619
$\hat{\lambda}$ per year (s.e.)	102 (11)	124 (53)
Jump share of variance	71%	51%

- Standardising removes most kurtosis, yet the model still reports a “jump” about every other day: $\hat{\lambda}$ measures fat tails, not discontinuities
- Interpretation: daily data do not identify the jump share; Lee–Mykland finds 0.87 jumps per year and 5-minute data give 2.8%, a different frequency and sample. Code: the chapter’s Quantlet (AI case study)

Your Turn: A Project Seed

- Question: how much of S&P 500 and Bitcoin variance is discontinuous, at which frequency, and are the detected “jumps” drift bursts?
- Replicate first: Merton jump share 71% (daily S&P 500), Lee–Mykland 0.87 jumps per year, bipower jump share 2.8% (SPY, 5 minutes, [Chapter 9](#))
- Deliverables
 - ▶ A pre-registration (hypotheses, frequencies, filters, tests, bootstrap design) dated before the first run
 - ▶ The jump share at 5, 15, 30 minutes and daily; the drift-burst statistic of [Christensen, Oomen & Renò \(2022\)](#) around April 2025; a bootstrap LR test of Heston (null, no jumps) against Bates
 - ▶ Code, a two-page report with every specification tried, and a referee-style critique of your own result
- Declare your AI use in `AI_USE.md` and the errors you caught in `AI_ERRORS.md`:
github.com/danpele/MFM/tree/main/project

Key Takeaways and Next: Chapter 12

- Brownian motion is the universal limit of random walks; its quadratic variation t drives Itô calculus
- Itô's lemma gives exact solutions (GBM, Ornstein–Uhlenbeck, Merton); simulate exactly, or with Euler–Maruyama and Milstein at a known error order
- Identification: infill pins down volatility; drift and mean-reversion speed need long spans, and $\hat{\kappa}$ is biased upwards
- Girsanov changes only the drift; Feynman–Kac turns a Q -expectation into a PDE; $(\text{VIX}/100)^2$ is a Q -expectation, affine in v_t
- In the fitted models jumps help with the tails and stochastic volatility with the clustering; volatility is rough and persistent, which one-factor models cannot produce
- Next, [Chapter 12](#): pricing options by replication and reading the volatility surface; code and data: github.com/danpele/MFM/tree/main/Quantlets/Ch_11

Appendix: From Brown to Bachelier, Einstein and Wiener



Louis Bachelier (1870–1946)

Photo: unknown author; public domain; Wikimedia Commons



Albert Einstein in 1905

Photo: Lucien Chavan / ETH Zürich (1905); public domain; Wikimedia Commons

- 1827: the botanist Robert Brown sees pollen particles move in an irregular zig-zag under the microscope
- Bachelier (1900): French government bond (rente) prices and their options follow a random walk in continuous time
 - ▶ The spread of the price grows with $\sqrt{t}$, five years before physics
- Einstein (1905): molecular collisions explain Brown's motion; the mean squared displacement is linear in t
- Wiener (1923): Brownian motion as a probability measure on continuous paths, the **Wiener process**

Appendix: Itô, Samuelson, Black, Scholes and Merton



Kiyosi Itô (1915–2008)

Photo: Konrad Jacobs (1970); CC
BY-SA 2.0 de; Wikimedia Commons



Paul A. Samuelson (1915–2009)

Photo: Bernard Gotfryd
(1970–1975); public domain;
Wikimedia Commons

- Itô (1944): the stochastic integral and its chain rule, the main working tool of this chapter
- Osborne (1959): log prices, not prices, behave like Brownian motion; Samuelson (1965): geometric Brownian motion
- Black & Scholes (1973) and Merton (1973): an option can be replicated in continuous time
- Merton (1976) adds jumps, Heston (1993) random volatility; textbook treatment: Franke, Härdle & Hafner (2019), Part I

Appendix: A Crash in Pictures, Amsterdam 1987



Amsterdam Stock Exchange, 21 October 1987, after the crash of 19 October

Photo: Bart Molendijk / Anefo (1987); CC0; Wikimedia Commons

- 19 October 1987: the Dow Jones fell 22.6% in one day, about 20 standard deviations under GBM with 18% volatility
- Under a Normal distribution such a day would not be expected once in the age of the universe: prices gap, and continuous models need jumps or fat-tailed innovations
- A gap cannot be hedged by continuous trading, the argument of [Merton \(1976\)](#)

Appendix: Data Cleaning and Technical Conditions

- Data: equity indices on weekdays, stale quotes removed (unchanged close and no intraday range), genuine zero returns kept; Bitcoin 7 days a week
- Itô integral: left-point sums converge for regular integrands (e.g. continuous and adapted); a general H is reached by approximation with simple processes
 - ▶ If only $\int_0^T H_t^2 dt < \infty$ almost surely, $\int H dW$ is merely a local martingale and its mean need not be zero
- SDE: existence and uniqueness of a strong solution if a and b are Lipschitz in x with at most linear growth [Kloeden & Platen \(1992\)](#)
 - ▶ Square-root coefficients (CIR, Heston) are not Lipschitz at zero, yet have unique strong solutions (Yamada–Watanabe conditions)
- Donsker: convergence in distribution in the space of càdlàg paths with the Skorokhod topology [Billingsley \(1999\)](#)

Appendix: Short Rates, More Numbers

- DTB3, 1954–2026: the rate ranged from -0.05% (2020-03-26) to 17.1% (1980-12-11); 4.3% of the days were below 0.05%
 - ▶ The fitted Vasicek model gives a 6.1% long-run probability of a negative rate
 - ▶ 2008–2026: 13 exact zeros and 8 negative quotes, outside the domain of the CIR and CKLS likelihoods; 17% of the days form a near-zero regime below 0.05%
- CKLS tests on γ , 1954–2007, HAC Wald with 11 (daily) and 6 (monthly) lags
 - ▶ $\gamma = 0$ rejected: daily 545, monthly 16; $\gamma = 1/2$ rejected daily (67.0; 22.1 with 55 lags), not monthly (0.5 against 3.84)
 - ▶ The Gaussian LR for $\gamma = 1/2$ rejects monthly (10.1): it ignores volatility clustering; [Chan, Karolyi, Longstaff & Sanders \(1992\)](#) found $\hat{\gamma} \approx 1.5$ for 1964–1989
- Vasicek bond with the Section 6 estimates, $\lambda = 0$ and $r = 3.99\%$: 10-year yield 4.05% ; the 10-year Treasury yield (FRED DGS10) on 2026-09-18: 5.01%
 - ▶ The gap is a term premium ($\lambda \neq 0$) plus misspecification
- Nonparametric estimation: [Stanton \(1997\)](#) also estimates the market price of interest-rate risk; the specification test of [Aït-Sahalia \(1996\)](#) over-rejects with persistent data [Pritsker \(1998\)](#)

Appendix: The VIX Correction by κ^Q and the Full Model Comparison

κ^Q	1	2.5	5	10
b	0.96	0.90	0.82	0.68
Local correction $\hat{\xi}/b$	0.58	0.62	0.68	0.82

	Data	GBM	Merton	Heston
Excess kurtosis	10.88	0.00	3.66	3.30
Skewness	-0.36	0.00	-0.32	0.00
Hill statistic	2.56	4.64	2.70	2.95
ACF of $ r $, lag 1	0.27	0.00	0.00	0.28
ACF of $ r $, sum of lags 1–20	5.32	0.00	0.00	4.78

- $\hat{\xi}/b$ is exact only at $v_t = \theta^Q$. Model columns: medians of 100 simulations; Hill statistic of the data 2.97, 2.56, 2.04 for $k = 2.5\%$, 5% , 10% of the returns, a tail index only under regular (Pareto-type) variation
- A Heston model with a lower Feller ratio could reach the kurtosis: short-horizon excess kurtosis $\approx 3\xi^2/(2\kappa\theta)$

Appendix: Further Reading

- Heavier tails: Lévy processes and α -stable distributions [Pele \(2014\)](#)
 - ▶ Crypto-assets form a distinct class by the shape of their return distributions [Pele et al. \(2023\)](#)
 - ▶ High-frequency information about Bitcoin volatility and tail risk [Pele & Mazurencu-Marinescu-Pele \(2019\)](#)
- Stochastic calculus: [Shreve \(2004\)](#), Chapters 5–6; [Franke, Härdle & Hafner \(2019\)](#), chapters on the Wiener process and stochastic integration
- Exercises with solutions on all models of this chapter: [Borak, Härdle & López-Cabrera \(2013\)](#); the Heston model: [Čížek, Härdle & Weron \(2011\)](#); applications: [Härdle, Chen & Overbeck \(2017\)](#)
- Monte Carlo methods: [Glasserman \(2003\)](#); numerical SDEs: [Kloeden & Platen \(1992\)](#)

Appendix: Case Study, Model, Data and Method

- **Model:** $\sigma_t = \xi e^{X_t}$ (eq. (1.3)), $X_t = \int_{-\infty}^t g(t-s) v_s dW_s$, a Brownian semistationary (BSS) process with kernel g and volatility-of-volatility process v_s (eq. (1.5)); the forecasts take $v_s \equiv 1$, a Gaussian X (Section 5)
 - ▶ Gamma kernel $g(x) = x^\alpha e^{-\lambda x}$ (Example 1.2): roughness α , short memory with exponential decay at rate λ
 - ▶ Power kernel $g(x) = x^\alpha (1+x)^{-\gamma-\alpha}$ (eq. (1.8)): $\beta = \gamma$ if $\gamma > 1$, $\beta = 2\gamma - 1$ if $\gamma \in (1/2, 1)$ (Remark 1.2); Gaussian alternative: the Cauchy class, $\rho(h) = (1 + |h|^{2\alpha+1})^{-\beta/(2\alpha+1)}$ (Section 1.1.1)
- **Step 1,** α from the variogram $\hat{\gamma}_2(k)$ of $\log \hat{\sigma}$: OLS of $\log \hat{\gamma}_2(k)$ on $\log k$, $k = 1, \dots, 6$, $\hat{\alpha} = (\hat{a}_1 - 1)/2$ (eq. (2.1))
 - ▶ Noise-robust NLLS: $\hat{\gamma}_2(k) = b_0 + b_1(k\Delta)^{2\alpha+1}$, the constant absorbs the measurement noise; average the estimates obtained with $m = 10, \dots, 20$ lags (Sections 2.1 and 3.1)
 - ▶ **Step 2:** β , γ or λ by matching the model ACF to the empirical one on the first $\lceil n^{1/3} \rceil$ lags (Sections 2.2 and 4.1)
- **Data:** E-mini S&P 500 futures, every trade, 2011–2014 (996 days), 09:30–16:00 New York time; $\hat{\sigma}_t^2 =$ pre-averaged bipower variation over Δ , divided by Δ , for Δ from 10 minutes to 1 day (Sections 4 and 4.1)
 - ▶ Intraday seasonality removed with the flexible Fourier form of [Andersen & Bollerslev \(1997\)](#); daily data on 1944 liquid US stocks, 2003–2013 (Sections 4.1 and 4.2)
- **Forecasts:** variance summed over the next $h = 1, 2, 5, 10, 20$ steps, rolling window of 200 observations, out of sample 2 January 2013 – 31 December 2014 (Section 5.1.1)
 - ▶ Rivals: random walk, AR(5), AR(10), ARMA(1,1), log-HAR (heterogeneous autoregressive model for log volatility) [Corsi \(2009\)](#), ARFIMA(0, d ,0), RFSV; losses MSE and QLIKE; 90% and 75% MCS [Hansen, Lunde & Nason \(2011\)](#)

Appendix: Case Study, Estimators and Forecast Losses

- **Variogram** of $x_t = \log \hat{\sigma}_t$: $\hat{\gamma}_2(k) = \frac{1}{n-k} \sum_{t=1}^{n-k} (x_{t+k} - x_t)^2$; for small k , $\gamma_2(k) \approx c k^{2\alpha+1}$ (Section 2.1)
 - ▶ **Bipower variation** of the M returns r_j in a bin: $BV = \frac{\pi}{2} \sum_{j=2}^M |r_j| |r_{j-1}|$, robust to jumps [Barndorff-Nielsen & Shephard \(2004\)](#); the pre-averaged version first averages returns over short windows to damp microstructure noise (Appendix A of the paper)
- **Forecast object**: $FO_t(\Delta, h) = \sum_{k=1}^h BV_{t+k\Delta}$ (eq. (5.3)); for a Gaussian log model, $\mathbb{E}[e^{X_{t+\Delta}} | \mathcal{F}_t] = \exp(m + s^2/2)$, with m, s^2 the conditional mean and variance (eq. (5.2))
 - ▶ Losses over the out-of-sample origins: $MSE = \text{mean}(\widehat{FO}_t - FO_t)^2$, $QLIKE = \text{mean}(\log \widehat{FO}_t + FO_t/\widehat{FO}_t)$ (Section 5.1.1)
- **Model confidence set (MCS)** [Hansen, Lunde & Nason \(2011\)](#): a set of models that contains the best model with probability 90% (or 75%)
 - ▶ Membership is not a ranking: the models inside the MCS cannot be told apart at that level

Appendix: Case Study, Results in the Paper

log σ , E-mini S&P 500 (Table 3, Panel A)	$\hat{\alpha}_{OLS}$	$\hat{\alpha}_{NLLS}^*$	$\hat{\beta}_{Cauchy}^*$	$\hat{\lambda}_{Gamma}^*$
$\Delta = 10$ minutes	−0.38	−0.38	0.18	0.02
$\Delta = 65$ minutes	−0.30	−0.35	0.18	0.02
$\Delta = 1$ day	−0.30	−0.33	0.18	0.02

- Rough at every scale: $\hat{\alpha}$ between −0.38 and −0.30 from 10 minutes to 1 day (Table 3, Panel A; * = noise-robust): log-volatility is far rougher than Brownian motion ($\alpha = 0$)
 - And persistent: $\hat{\beta}_{Cauchy}^*$ between 0.17 and 0.18 at every Δ , below 1, and $\hat{\lambda}^*$ between 0.02 and 0.03 (Table 3, Panel A): long memory together with roughness
 - 1 944 US stocks: $\hat{\alpha}^*$ mostly between −0.45 and −0.30, $\hat{\beta}^*$ mostly between 0 and 0.4, no systematic difference across sectors (Figures 4 and 5): both features look universal
- Noise can fake roughness: a simulated $\alpha = 0$ observed with noise-to-signal ratio 0.3 at $\Delta = 0.1$ gives a mean OLS estimate of −0.47 but a mean NLLS estimate of 0.01 (Table 1, Panel C)
- 15-minute steps, $h = 20$: MSE ($\times 10^8$) 0.038 for log-HAR, 0.020 for RFSV, 0.019 for Power-BSS (Table 4, Panel A); daily steps, $h = 1$: RFSV is best, 0.493 ($\times 10^9$) (Table 5, Panel B): modelling persistence pays intraday and at long horizons

Appendix: Case Study, Intraday Application, Limits and Open Questions

- **Intraday application** (illustrative, not an exact replication) on the SPY 5-minute bars of the course data: $\hat{\sigma}_t^2$ = bipower variation of the 5-minute returns in each bin of $\Delta = 30$ or 65 minutes, divided by Δ and by the flexible Fourier seasonal factor Andersen & Bollerslev (1997)
 - ▶ OLS and NLLS $\hat{\alpha}$, Cauchy $\hat{\beta}^*$ and Gamma-BSS $\hat{\lambda}^*$ against Table 3, Panel A; log-HAR, RFSV and BSS forecasts, MSE, QLIKE and MCS against Table 5
 - ▶ Data limit: bars instead of every trade, so the pre-averaged estimator (Appendix A of the paper) is not feasible and the noise is larger; only 13 returns per 65-minute bin
- **Limits stated by the authors**
 - ▶ Realised measures are noisy estimates of integrated variance: the conclusions concern them, not the latent spot volatility (Sections 4.1, 4.2 and 6)
 - ▶ No rule for the NLLS bandwidth m (Sections 2.1 and 3.1); NLLS is too costly for rolling forecasts, which use the OLS $\hat{\alpha}$ (Section 5.1.1)
 - ▶ Forecasts assume a Gaussian X ; a non-Gaussian volatility of volatility is left out of scope (Section 5; footnote 3)
- **Open question:** why does long memory improve even short-horizon forecasts? The authors leave it to further research (Section 5.1.2)
 - ▶ Rolling $\hat{\alpha}$ (65 minutes, 2005–2014, median -0.36) is less rough in 2007–2008 and 2011 (Figure 3), and $\hat{\alpha}$ rises with log volume, slope 0.0091 (Figure 6): does Bitcoin (5-minute bars), a market that never closes, fit this pattern?

Appendix: The Discovery Loop, Reusable Prompts (1/2)

- Tools: an AI assistant (an LLM such as Claude, ChatGPT, Gemini or GitHub Copilot); literature search: [Semantic Scholar](#), [Elicit](#), [Google Scholar](#)
- 1. **Literature**: tabulate estimates by frequency, then check every DOI on [Crossref](#) (the title must match)
 - ▶ Tabulate published estimates of the jump share of equity-index variance by sampling frequency (daily, 5-minute, tick) and method; give the exact number, the table it comes from, and the DOI.
- 2. **Competing hypotheses**, with distinct testable implications
 - ▶ For three processes that give a daily excess kurtosis near 11 (Merton jumps, Heston volatility, drift bursts), derive the implications for returns standardised by ex-ante volatility, for bipower versus realised variance, and for the drift just before extreme moves.
- 3. **Data and code**: replicate the chapter's estimate first
 - ▶ Write Python that reads the S&P 500 daily closes of the course and fits the Merton model by maximum likelihood (Poisson mixture truncated at 12 jumps, several starting points). Target: $\hat{\lambda} = 102$ per year, jump share 71%

Appendix: The Discovery Loop, Reusable Prompts (2/2)

- 4. **Robustness**, pre-registered
 - ▶ Write a pre-registered plan: standardise returns by EWMA and by GARCH volatility known at $t-1$; split 1990-2007 and 2008-2026; test Heston (no jumps, the null) against Bates (SVJ); obtain LR critical values by parametric bootstrap under the Heston null.
- 5. **Critique**: ask the assistant to attack the inference
 - ▶ Act as a referee: why is the LR test of $\lambda = 0$ non-standard, what is the correct reference distribution, and could a Poisson mixture with frequent small jumps simply be a fat-tailed innovation?
- 6. **Report**: all frequencies and filters, bootstrap critical values, code and the AI log

Appendix: Further Study on Quantinar

Advanced courses on quantinar.com

- ☐  Applied Computational Finance
- ☐  XFG Advanced Methods in Quantitative Finance

Code and data

- ☐ Quantlets: github.com/danpele/MFM/tree/main/Quantlets/Ch_11



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